

# Westchester County Data Platform

## Executive Summary

Arcanum Research Initiative

October 2025

### Overview

The Westchester County Data Platform is a comprehensive data analysis and visualization system providing evidence-based insights for county planning, policy analysis, and research applications. The platform integrates multiple open data sources into a unified, accessible web interface with professional-grade dashboards and Druck-compliant data outputs.

### Key Findings

#### Demographics

- **Population:** Comprehensive demographic data collected for Westchester County covering 241 census tracts
- **Income Analysis:** Median household income, per capita income, and poverty statistics available at county and tract levels
- **Housing Metrics:** Total housing units, occupancy rates, median home values, and rental statistics integrated

#### Transit Accessibility

- **Station Coverage:** 56 Metro-North Railroad stations mapped within Westchester County
- **Accessibility:** Wheelchair accessibility data available for all stations
- **Geographic Reach:** 1-mile walkable radius coverage analysis completed

#### Data Integration

- **Sources Integrated:** 3 primary data sources (Census Bureau, Metro-North GTFS, NY State Open Data)
- **Records Processed:** 1,611 crime records, 330 health facilities, 56 transit stations
- **Geographic Scope:** Full Westchester County coverage with precise coordinate filtering

## Deliverables

### Data Outputs

- 4 Druck-compliant Excel files (ONE sheet per file)
- County-level demographics spreadsheet
- Census tract analysis (241 tracts)
- Metro-North station directory
- Transit accessibility metrics

### Web Platform

- Interactive React/TypeScript frontend application
- 7 professional dashboards:
  - Overview Dashboard
  - Demographics Dashboard
  - Transit Accessibility Dashboard
  - Property Tax Dashboard
  - County Budget Dashboard
  - Municipal Services Dashboard
  - Municipality Comparison Dashboard
- FastAPI backend with 8 RESTful endpoints
- Interactive mapping with Leaflet
- Statistical visualizations with Recharts

### Documentation

- Comprehensive methodology report
- Technical implementation guide
- API documentation
- Data source citations and references

## Technical Architecture

### Technology Stack:

- Backend: Python 3.11+, FastAPI, pandas, geopandas
- Frontend: React 18, TypeScript, Vite, Tailwind CSS

- Data Analysis: Jupyter Lab, openpyxl
- Deployment: Local-first architecture, Docker-ready

**Data Processing Pipeline:**

1. Automated API data collection
2. Geographic filtering and transformation
3. Druck-compliant Excel generation
4. Interactive web visualization

Applications

Primary Use Cases

- **County Planning:** Evidence-based decision making for infrastructure and development
- **Policy Analysis:** Data-driven insights for government officials
- **Academic Research:** Comprehensive dataset for demographic and economic studies
- **Public Access:** Transparent data visualization for county residents
- **Media Analysis:** Statistical foundation for journalistic reporting

Target Audience

- Westchester County Government officials and planners
- Municipal administrators and town managers
- Academic researchers and policy analysts
- Journalists and media organizations
- Engaged citizens and community organizations

Success Metrics

| Category         | Target          | Achievement    |
|------------------|-----------------|----------------|
| Data Sources     | 3+ integrated   | 3 (100%)       |
| Excel Compliance | Druck standards | 100% compliant |
| Dashboard Count  | 5-7 dashboards  | 7 (100%)       |
| API Endpoints    | 6+ endpoints    | 8 (133%)       |
| Documentation    | Complete        | 100%           |

## Next Steps

### Immediate Priorities

1. Deploy web platform to production server
2. Complete property tax data integration
3. Add multi-year historical trend analysis
4. Implement automated data refresh schedules

### Future Enhancements

- PostgreSQL + PostGIS database integration
- Advanced spatial analysis tools
- Real-time data streaming
- Mobile-responsive design optimization
- Export functionality for custom reports

## Conclusion

The Westchester County Data Platform successfully integrates comprehensive open data sources into a professional-grade analysis and visualization system. With full Druck compliance, operational dashboards, and extensive documentation, the platform provides a robust foundation for evidence-based decision making in county planning and policy analysis.

### For More Information:

- Methodology Report: `westchester_methodology_report.pdf`
- Technical Documentation: `Projects/Westchester/Technical/README.md`
- API Documentation: `http://localhost:8000/docs`